DESIGN BRIEF · MILESTONE 2, PHASE 1 · OCT 14, 2026

Query Module 2.0

A Query Module for the CAD suite where dispatchers and officers query state and national systems from one fast, accessible screen, read the answers where they ask, and where an implementer reshapes the screen without a code release.

THE PROBLEM

Answers under time pressure, on every site's own terms

A dispatcher or officer asks a state or national system about a plate, a person, a driver's license or a piece of property, and has seconds to read the answer back over the radio. The first release did that, but the verdict on it was blunt: it works, and it reads as a developer tool. Every site also wants different fields, rules, commands and result layouts, and a code release per site is not workable.

The target is an operations console: dense, calm, dark-first, one clear accent, nothing decorative. The design holds to four goals. The answer is readable at a glance. A serious result is hard to miss. The screens are built to work by keyboard. And the people who run a site can change what the screen shows, see what dispatchers will see, and publish it.

THREE ROLES

Who the screens are for

Dispatcher

DESK · 36 PX CONTROLS

- Two panes: the query panel and Requests this shift
- Quick access to the five common queries, Alt+1 to Alt+5
- Form or one-line terminal command, one shared draft
- Results read in place, in the request that asked

Officer

VEHICLE LAPTOP · 1024 x 768

- One card at touch density, targets of 48 px or more
- Large query tiles and a 64 px Run query
- Night and red-shift themes one tap away
- No muted text for anything that must be read; detail is one tap behind the summary

Implementer and administrator

BUILDER · NO JSON NEEDED

- Shapes fields, rules, commands, result cards and keywords in plain language
- Previews on the dispatcher's own screen, then publishes and can roll back
- The implementer edits site configuration only
- The administrator does the same and also manages users, roles and sessions

DESIGN PRINCIPLES

Eight rules every screen is held to

1 Configuration, not code

Fields, rules, commands, result cards and keywords are site configuration. A change is published, not released.

2 The preview is the live form

The builder previews with the same renderer dispatchers use. What the implementer sees is what goes live.

3 Accessible by design

Built for keyboard and screen reader use from the start, with a visible focus ring and body text at 7:1 contrast or better in every theme. Browser tests drive the main dispatcher and officer flows by keyboard, run an automated accessibility scan on the main screens, and check the query screen at 200% zoom and 320 px wide.

4 Night and red shift

Day, night and a red-shift mode that protects night vision. Each has its own focus ring and contrast proof.

5 Severity is never color alone

A serious result carries an icon, the word, a visible marker, a hidden spoken severity and a 4 px edge, as well as its color.

6 Read-back text in monospace

Plates, VINs, serial numbers, commands and reference IDs are set in a monospace face so they can be read back character by character.

7 One draft for form and terminal

A command echo under the form shows the terminal command being built. Switching modes keeps what was typed.

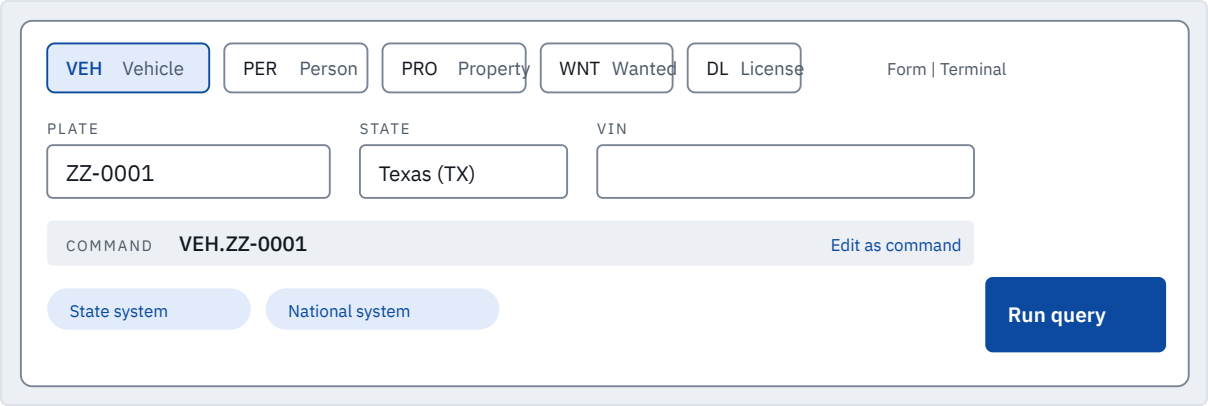
8 Focus is never lost

Incoming results, status changes and published config are built not to move focus or rebuild the control the user is on.

DIRECTION

An operations console

Four surface steps carry the hierarchy, so panels need few borders. One accent marks what is selected or actionable. Severity colors are reserved for responses and badges. The signature element is the command echo: under every query form a monospace strip shows the terminal command the form is building, live as the user types, with an Edit as command action. It teaches the terminal syntax and makes the form and the terminal one thing.

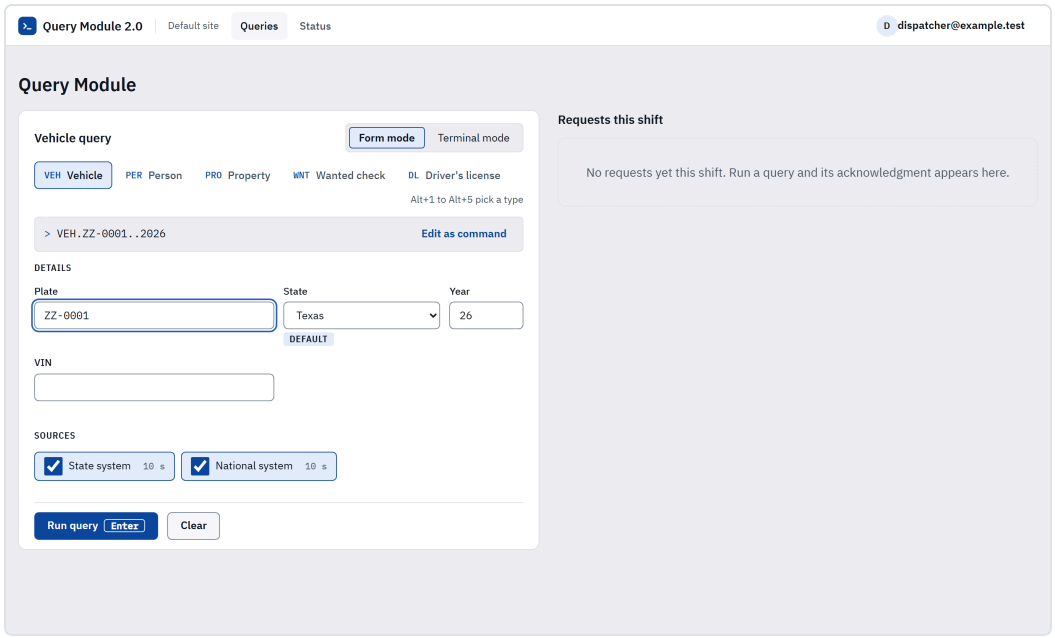


The dispatcher's query panel: quick access, the fields, the command echo, the sources and one primary action. Redrawn from the design; the live screen is on the next pages.

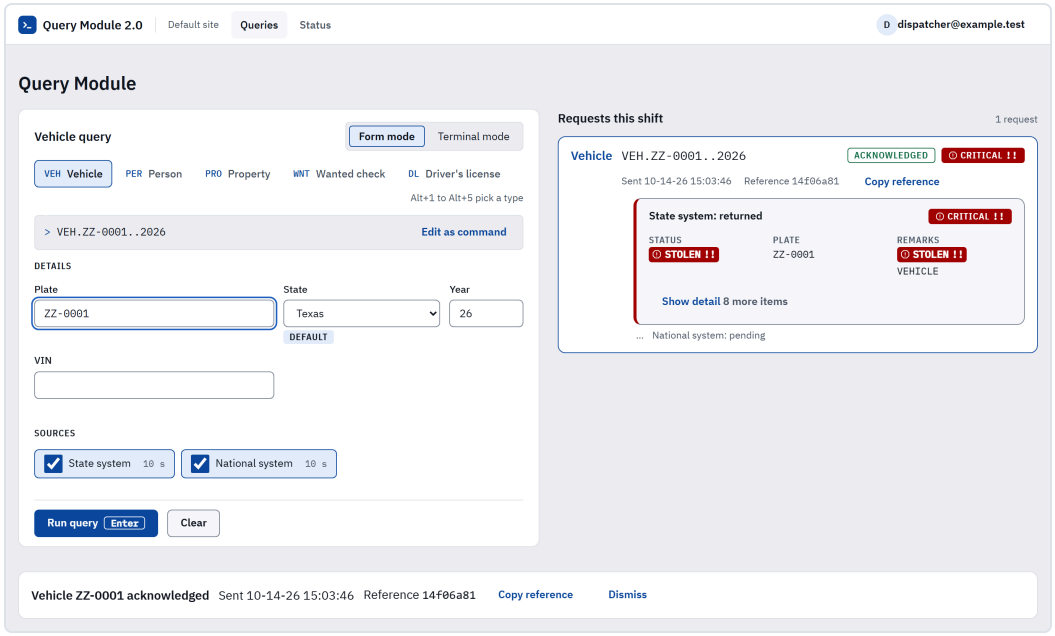
KEY FLOW 1

Ask, acknowledge, watch each source answer

Run query records the request and answers at once with an acknowledgment: a row in Requests this shift with the send time and a short reference. Under it, one line per source reads pending, then settles on its own as that source answers. A slow or failed source never holds up another. The text carries the status; an icon is never the only signal, and a screen reader hears one summary after each change.



The query panel as built: Vehicle form with the command echo and sources.

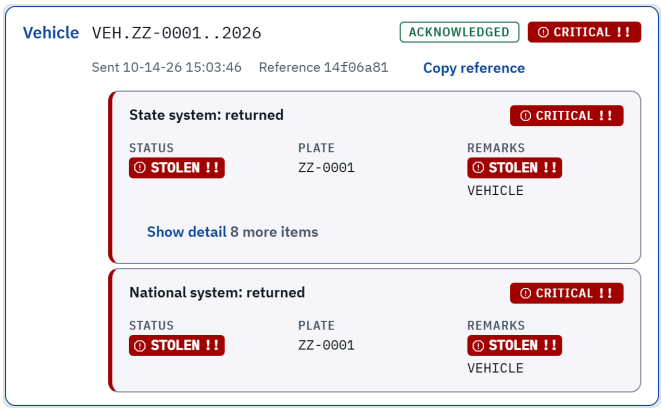


An acknowledged request with each source settling on its own.

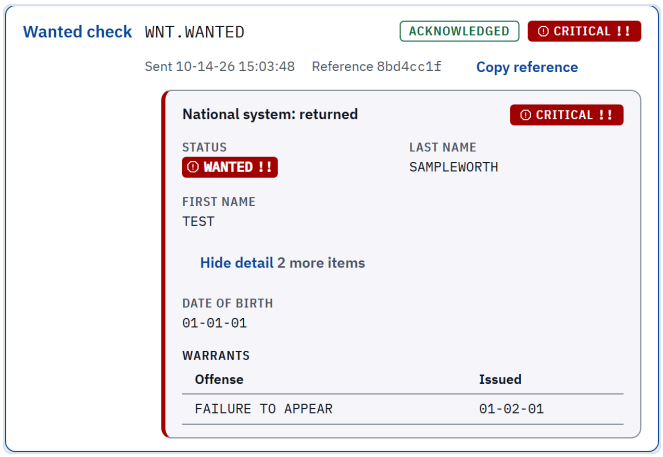
KEY FLOW 2

Results where they were asked: the result card

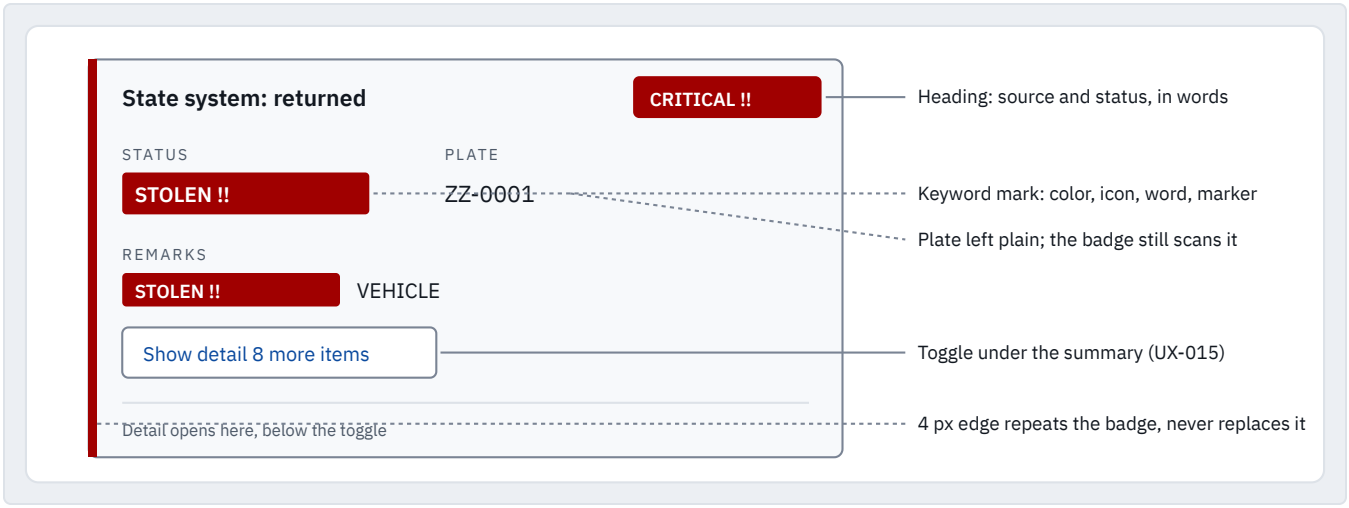
Each source that returns draws a card inside the request. The card is a view over the answer, configured by the site. The summary comes first and is short; Show detail opens the rest below it without moving the toggle. A serious word in the answer is marked, and a badge on the card, on the request and in the announcement carries the highest severity, even when the word sits in a row nobody opened.



A critical result: badge, mark on the word, 4 px edge, plate left plain.



Detail open: a warrants table in its own scroll box.



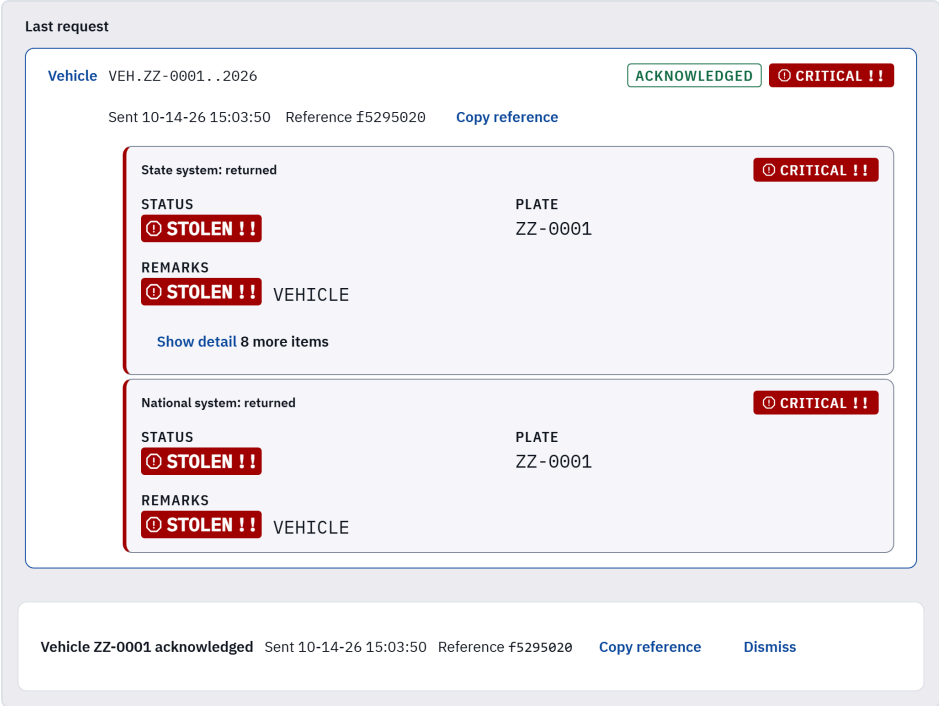
Anatomy of a card. Redrawn from the design; every severity cue has a non-color twin.

When no card is set up for a result, the card says so and shows every field under its path in the answer, so nothing is hidden. A source that is pending, timed out, failed or not run is one status line with its reason in words, never a card. A nested query that was skipped says why in a sentence.

KEY FLOW 3

The same card on the vehicle laptop

The officer layout uses the same renderer at touch density. Summary values are 20 px monospace, labels are 16 px in the body color, touch targets are 48 px or more, and detail stays one tap behind the summary.

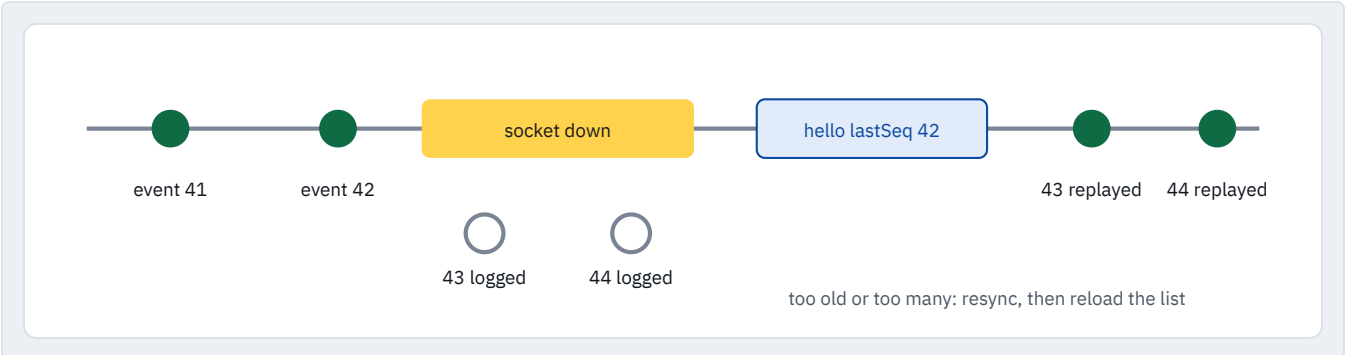


Officer layout at 1024 x 768.

KEY FLOW 4

Nothing is lost when the connection is

Live status travels over a socket. Every event is also written to a per-user log, and the client remembers the last event it applied. After a drop or a reload it says where it was and the server replays what it missed. If it was away too long, the server says so and the client reloads the request list instead.

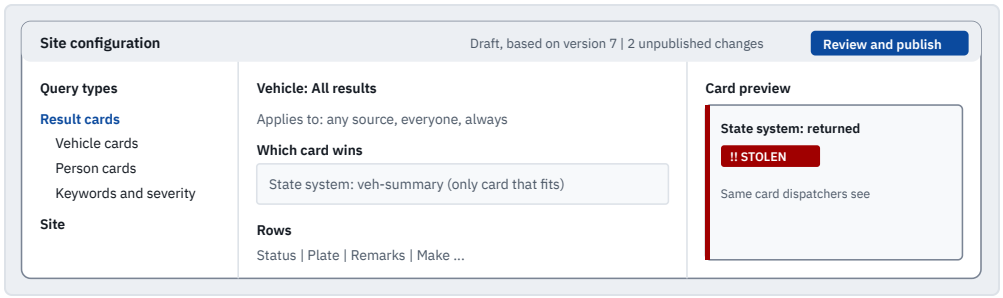


Replay by last sequence. At most 500 events from the last 24 hours; otherwise resync.

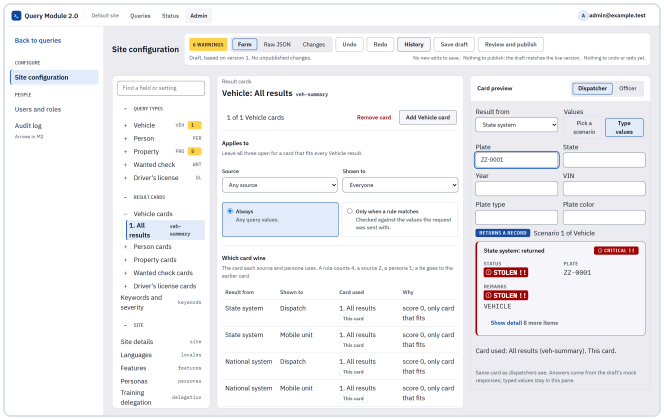
KEY FLOW 5

The builder: cards, keywords and Try a match

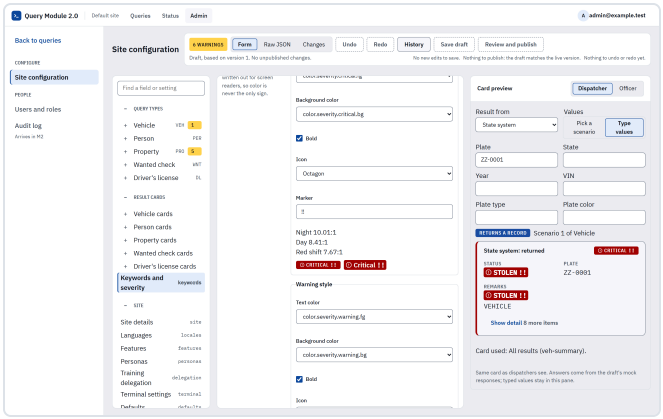
Implementers shape the result cards in the same builder that shapes the query forms: a tree on the left, an editor in the middle and a live preview on the right, with undo, Review and publish, history and roll back. A published change reaches open dispatcher screens within 15 seconds and redraws the cards already on screen, keeping what the user was typing.



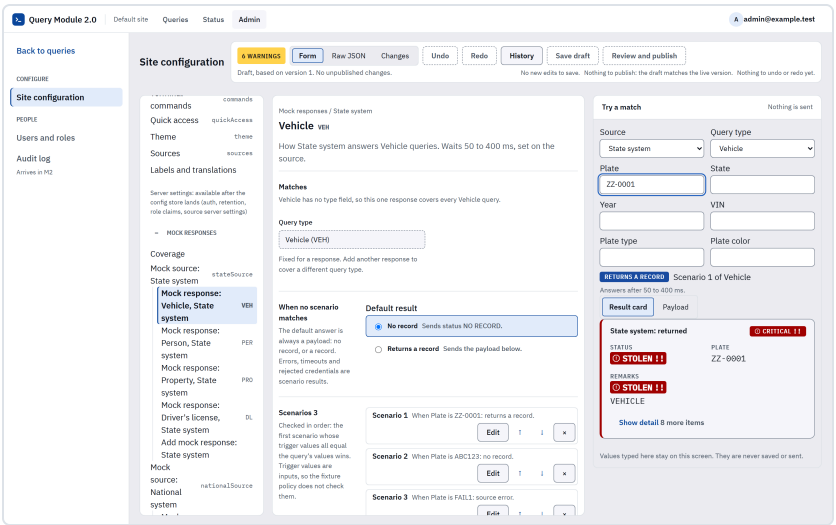
The builder with the card editor open. Redrawn from the design.



Card editor, Which card wins and the card preview.



Keywords and severity, with contrast per theme.



Try a match: type values, see which mock scenario answers. Nothing is sent or saved.

DESIGN DECISIONS

What was decided, and why

Each row is a decision the developer approved or a rule the product is held to. The visual system and the result-card design note are the sources; they carry the full text.

#	DECISION	WHY	SOURCE
1	A dark-first operations console with one accent and nothing decorative.	The first release read as a developer tool; an operator needs calm and density.	Visual system: Why, Direction
2	A command echo under every query form.	It teaches the terminal syntax and makes the form and the terminal one draft.	Visual system: Direction; FR-050 to FR-055
3	The night focus ring is amber; the red-shift ring is pale amber.	A red ring beside the red invalid edge would blur the two; amber stays visible on near-black.	Ruling 09-29-26 (1); E1
4	IBM Plex (Sans, Condensed, Mono), self-hosted.	One family with a monospace face for read-back text, served by the product itself.	Ruling 09-29-26 (2)
5	36 px dispatcher controls; officer targets of 48 px or more and a 64 px Run query.	Density for the desk, reach and certainty for a vehicle laptop.	Ruling 09-29-26 (3); spec 6.3
6	The mockup is the target; each capability ships in its own phase.	A control that cannot work yet is disabled with a visible reason instead of being hidden.	Ruling 09-29-26 (4); spec 6.2
7	Summary first; Show detail sits directly under the summary and opens below it.	The first read stays short, and the toggle and everything above it never move.	Mockup v5 rulings 1 and 2; UX-015
8	Severity is never color alone: icon, word, visible marker, spoken severity and a 4 px edge.	Color fails for some users and some displays; every cue has a twin.	Mockup v5 rulings 3 and 8; UX-010, UX-011
9	The severity badge scans the whole answer, not just the rows on the card.	A serious word in a detail-only row must still raise the badge and the announcement.	Spec 4.5; FR-060
10	Below 4.5:1 contrast blocks publish; 4.5 to 7:1 shows a hint; shipped styles are 7:1 or better.	An implementer cannot publish a style that fails the floor, and is nudged toward the better one.	Mockup v5 ruling 7
11	A skipped nested query says why, in a sentence that names the missing field.	A silent gap looks like a clean result.	Mockup v5 ruling 4
12	Announcements carry the query type, the typed field, counts, status and severity, never answer content.	Spoken and logged text must not leak what the answer said.	Mockup v5 ruling 12; spec 5.9, 6.6
13	The preview in the builder is the production card, fed by the draft.	What an implementer approves is exactly what dispatchers will see.	ADR-0011 (4)
14	A result with no card shows every field under its path, with no size cap.	Nothing is hidden by a missing configuration.	Mockup v5 ruling 5; spec 4.5
15	No Retry button in this phase.	Retry behavior belongs with the spec 5.7 work, so this phase leaves it out.	Mockup v5 ruling 9; spec 5.7
16	Mock data only, with a fixture policy that checks every mock payload against an allowlist of synthetic patterns.	The prototype must never hold or imitate real person, vehicle or property records.	Spec 5.4, 10.8

Sources: `docs/design/2026-09-29-visual-system.md` (rulings 09-29-26 and 10-05-26, and the result-card section with the rulings approved 10-13-26), `docs/design/2026-10-07-result-card-config.md`, ADR-0009, ADR-0010, ADR-0011, ADR-0013 and the requirements (FR-, UX-, SEC- IDs).

WHAT COMES NEXT

The same screens, with more behind them

M2 phase 2: audit

- Admin audit viewer and export
- Receipt metric and response timing
- Mapping paths checked against the mock answers on publish

M3: workflow and compliance

- Credentials and delegation for training officers
- Multi-source and nested queries, and delete from view
- Multi-factor sign-in

M4: mobile and host

- Mobile layouts and the CAD mobile app
- Host integration
- Real state and national connections come later, after the security controls are signed off

Prototype scope: the Query Module runs on mock data with canned responses. It never connects to real state or national systems and holds no real person, vehicle or property records. Every value shown in this brief is synthetic.

Status at M2 P1 (Oct 14, 2026): 251 merged changes, 403 work items, 13 architecture decisions, 5,010 automated tests and 267 browser scenarios.